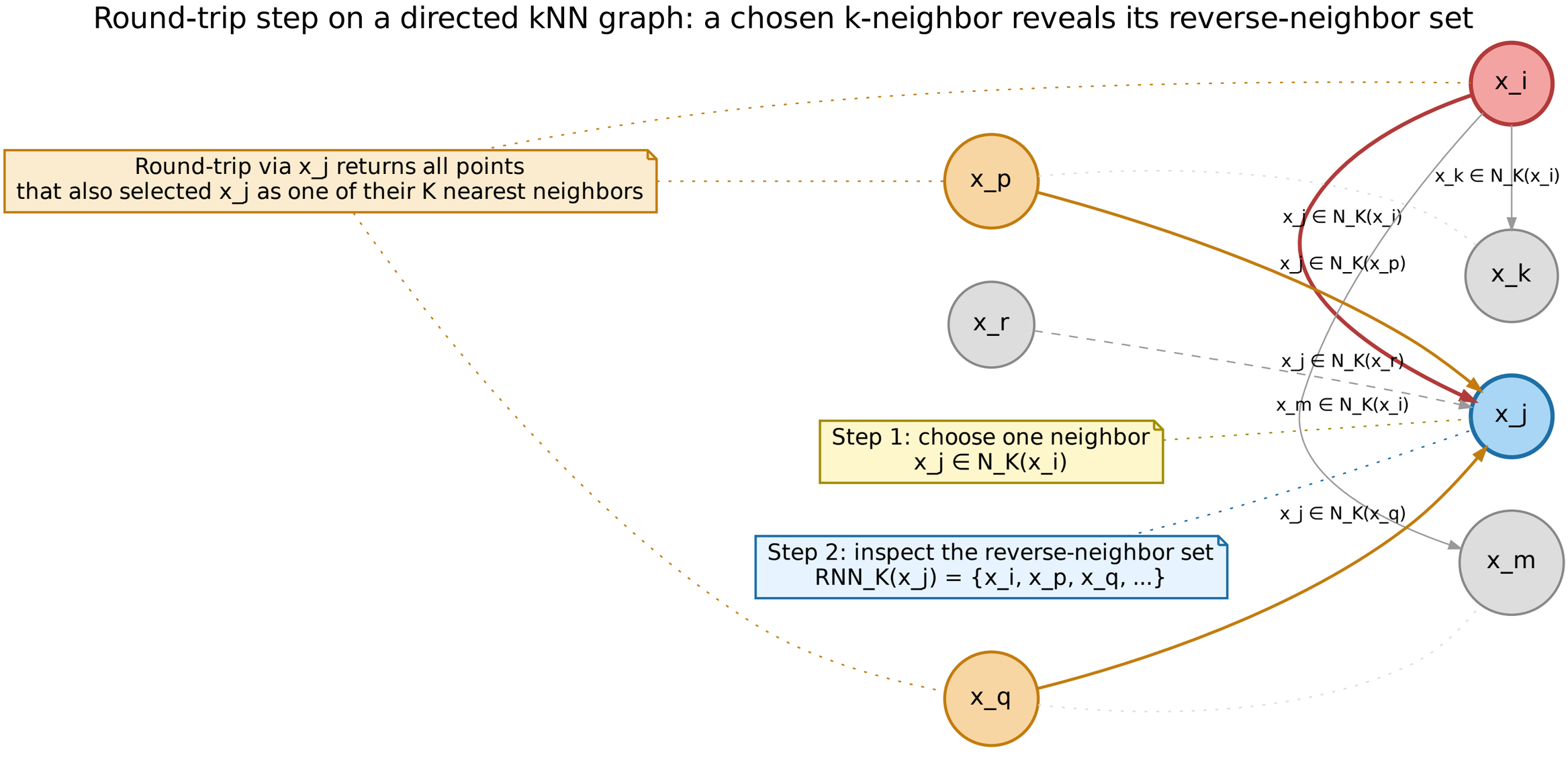
Round-trip step on a directed kNN graph: a chosen k-neighbor reveals its reverse-neighbor set
Round-trip via x_j returns all points
that also selected x_j as one of their K nearest neighbors
x_i
x_p
x_k ∈ N_K(x_i)
x_j ∈ N_K(x_i)
x_j ∈ N_K(x_p)
x_k
x_r
x_j ∈ N_K(x_r)
x_m ∈ N_K(x_i)
x_j
Step 1: choose one neighbor
x_j ∈ N_K(x_i)
x_j ∈ N_K(x_q)
Step 2: inspect the reverse-neighbor set
RNN_K(x_j) = {x_i, x_p, x_q, ...}
x_m
x_q